

Derivation of Choked Flow

Numbered Equations

Section 1

$$\rho A c = (\rho + d\rho) A (c - dv) \quad (1)$$

$$\rho dv = c d\rho \quad (2)$$

$$dp = \rho c dv \quad (3)$$

$$dp = c^2 d\rho \quad (4)$$

$$c^2 = \frac{dp}{d\rho} \quad (5)$$

Section 2

$$dq = du + p dv_s = 0 \quad (6)$$

$$0 = C_v dT + p dv_s \quad (7)$$

$$p dv_s + v_s dp = R dT \quad (8)$$

$$0 = (C_v + R) p dv_s + C_v v_s dp \quad (9)$$

$$0 = C_p p dv_s + C_v v_s dp \quad (10)$$

$$\frac{dp}{p} + \gamma \frac{dv_s}{v_s} = 0 \quad (11)$$

$$p v_s^\gamma = \text{const} \quad (12)$$

$$p \rho^{-\gamma} = \text{const} \iff p = K \rho^\gamma \quad (13)$$

Section 3

$$\frac{dp}{d\rho} = K \gamma \rho^{\gamma-1} = \gamma \frac{p}{\rho} \quad (14)$$

$$c = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\gamma R T} \quad (15)$$

Section 4

$$M = \frac{v}{c} \quad (16)$$

Section 5

$$\frac{d\rho}{\rho} + \frac{dA}{A} + \frac{dv}{v} = 0 \quad (17)$$

$$dp + \rho v dv = 0 \quad (18)$$

$$dp = c^2 d\rho \quad (19)$$

$$\frac{d\rho}{\rho} = -M^2 \frac{dv}{v} \quad (20)$$

$$\frac{dA}{A} = (M^2 - 1) \frac{dv}{v} \quad (21)$$

Section 6

$$h_0 = h + \frac{v^2}{2} \quad (22)$$

$$C_p T_0 = C_p T + \frac{v^2}{2} \quad (23)$$

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2 \quad (24)$$

$$\frac{p_0}{p} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\gamma/(\gamma-1)} \quad (25)$$

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{1/(\gamma-1)} \quad (26)$$

Section 7

$$\frac{p_0}{p^*} = \left(\frac{\gamma + 1}{2}\right)^{\gamma/(\gamma-1)} \quad (27)$$

$$\frac{p^*}{p_0} = \left(\frac{2}{\gamma + 1}\right)^{\gamma/(\gamma-1)} \quad (28)$$

$$\frac{p^*}{p_0} = 0.528 \quad (\gamma = 1.4) \quad (29)$$

$$\frac{T^*}{T_0} = \frac{2}{\gamma + 1} \quad (30)$$

$$\frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma + 1}\right)^{1/(\gamma-1)} \quad (31)$$

Section 8

$$\dot{m} = \rho^* A^* v^* \quad (32)$$

$$v^* = \sqrt{\gamma R T^*} = \sqrt{\gamma R T_0 \frac{2}{\gamma + 1}} \quad (33)$$

$$\rho^* = \rho_0 \left(\frac{2}{\gamma + 1} \right)^{1/(\gamma - 1)} \quad (34)$$

$$\dot{m}_{\max} = A^* p_0 \sqrt{\frac{\gamma}{R T_0} \left(\frac{2}{\gamma + 1} \right)^{(\gamma + 1)/(\gamma - 1)}} \quad (35)$$